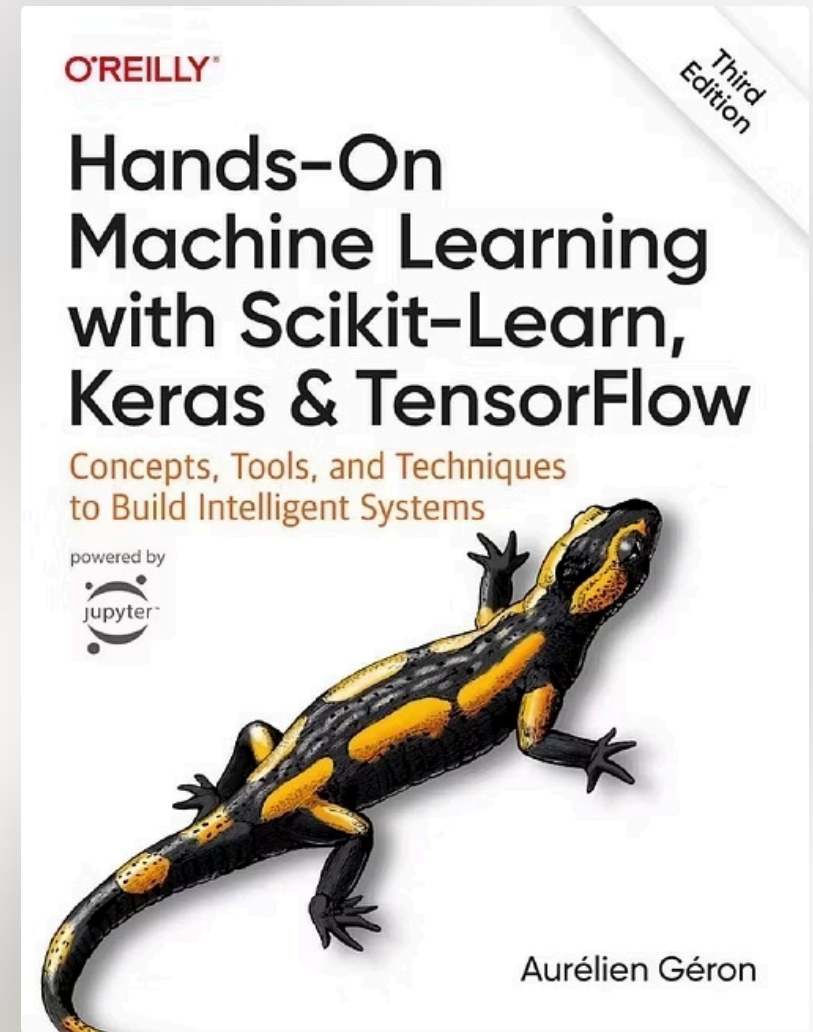


Machine Learning

Lessons based on the Textbook "Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow" by Aurélien Géron, 2023.

RP by Rodrigo Martins Pagliares

Last update: 07/01/2025



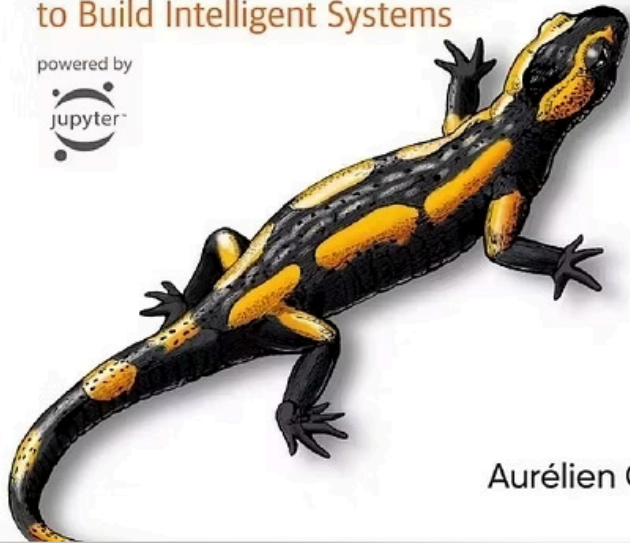
O'REILLY®

Third
Edition

Hands-On Machine Learning with Scikit-Learn, Keras & TensorFlow

Concepts, Tools, and Techniques
to Build Intelligent Systems

powered by



Aurélien Géron

Machine Learning

Chapter 1 – The Machine Learning Landscape



Agenda for Chapter 1

1. What Is Machine Learning?
2. Why Use Machine Learning?
- 3. Examples of Applications**
4. Types of Machine Learning Systems
5. Main Challenges of Machine Learning
6. Testing and Validating

Agenda for Chapter 1

1. What Is Machine Learning?
2. Why Use Machine Learning?
3. **Examples of Applications**
4. **Types of Machine Learning Systems**
 - Classification criteria #1: Supervision during training
 - Supervised, Unsupervised, Semi-Supervised, Self-Supervised, and Reinforcement Learning
 - Classification criteria #2: Approach to Learning Method
 - Batch and Online (Incremental) Learning
 - Classification criteria #3: Learning Method
 - Instance-Based and Model-Based Learning
5. **Main Challenges of Machine Learning**
 - Insufficient Quantity of Training Data
 - Non-representative Training Data
 - Poor-Quality Data
 - Irrelevant Features
 - Overfitting the Training Data
 - Underfitting the Training Data
 - Stepping Back
6. **Testing and Validating**
 - Hyperparameter Tuning and Model Selection
 - Data Mismatch



Examples of Applications

Some Applications of ML



Image Classification

Classifying **products** on a **production line** (using **CNNs** or **transformers**).



Tumor Detection

Semantic **image segmentation** for detecting tumors in medical scans (using **CNNs** or **transformers**).



Text Classification

Classifying **news articles**, **flagging** offensive **content**, and **summarizing** long **documents** automatically (using **NLP models**).



Some Applications of ML (cont.)



Regression

Predicting future revenues or property prices
(using linear regression, random forests).



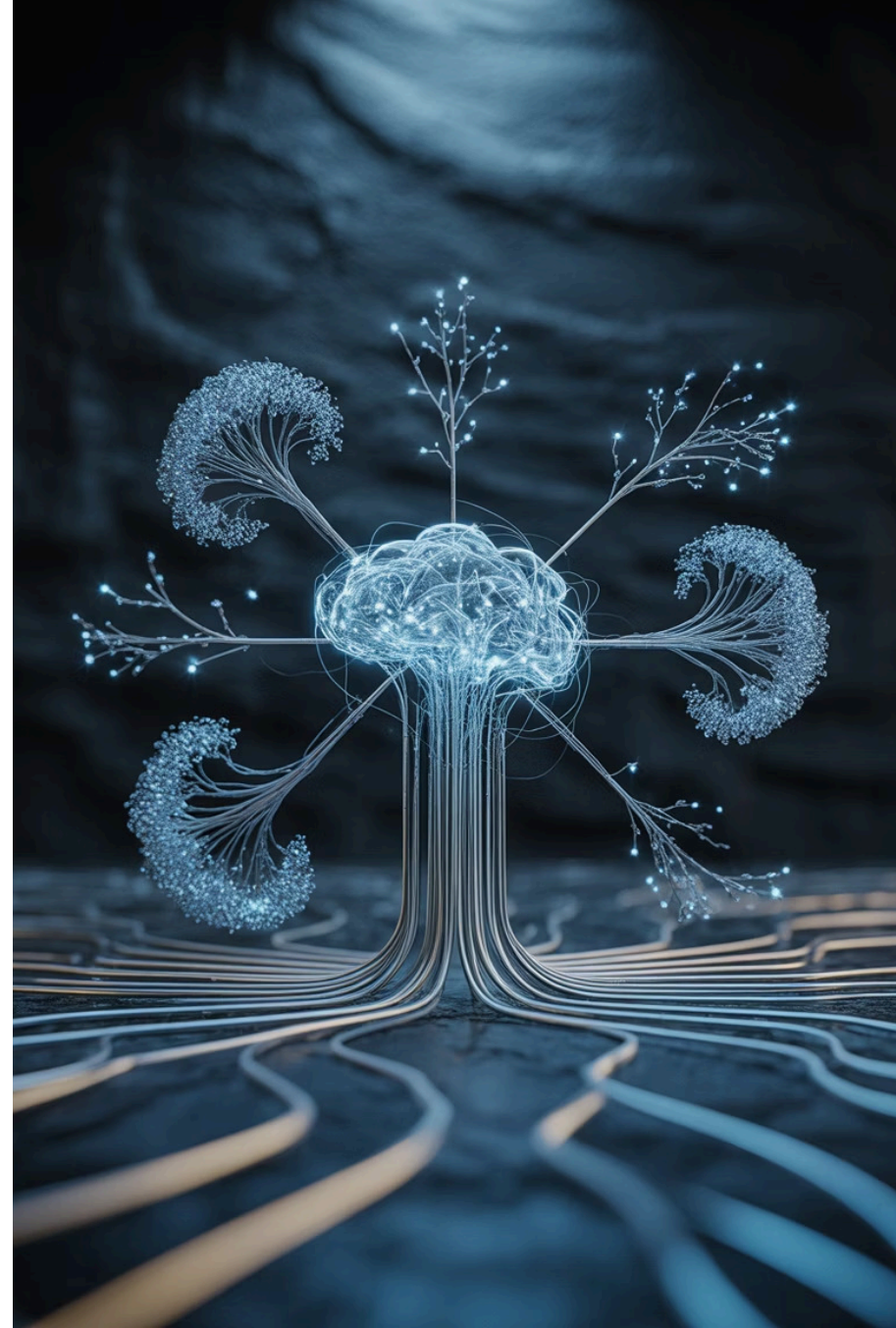
Anomaly Detection

Identifying fraudulent credit card transactions
(using isolation forests, autoencoders).



See Textbook for more examples of ML applications

Next: Types of Machine Learning Systems

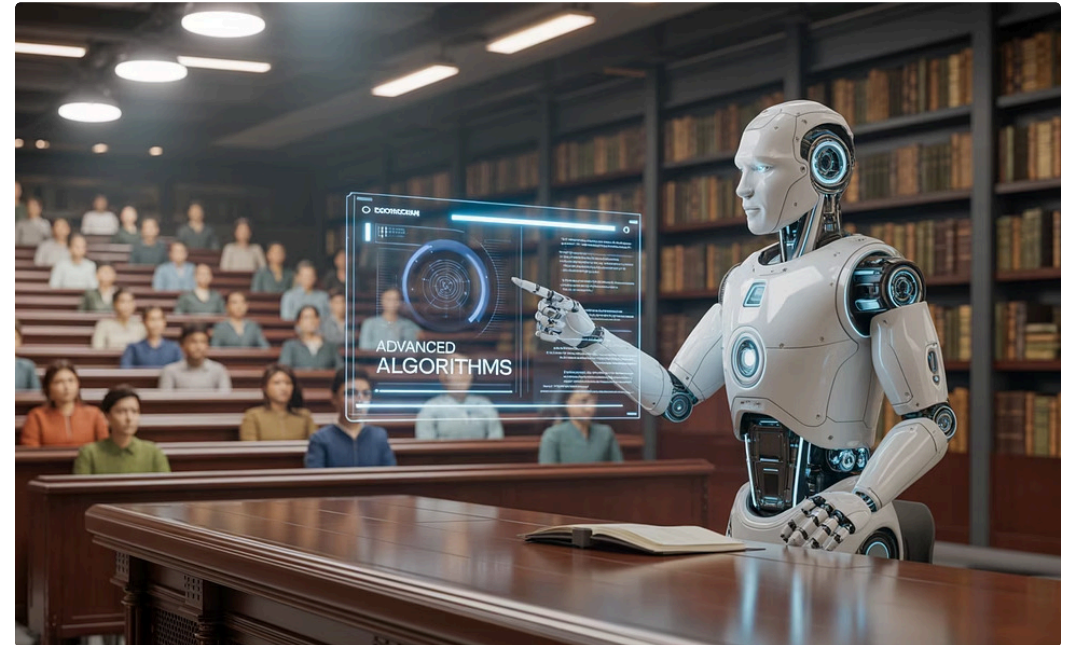


Q&A



Open Floor

Open floor for discussion or clarifications



Discussion

Share your thoughts and questions about AI concepts